

ChinookAnalytics: SQL Analytical Data Platform

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Abstract

ChinookAnalytics is a SQL-based analytical data platform designed to transform raw transactional music-store data into validated, structured, and business-aligned analytical assets.

The project implements a full layered pipeline (staging, core, marts), enforces relational integrity, formalizes analytical objectives, and delivers decision-oriented outputs supported by reproducible validation.

This document presents the architectural design, modeling strategy, analytical framework, and executive-level findings of the platform.

1 Product Scope and Objectives

ChinookAnalytics is designed as a complete analytical engineering system. Its primary objectives are:

- Ingest raw relational data in a controlled environment,
- Enforce referential and domain integrity at the core layer,
- Translate stakeholder questions into formal analytical models,
- Materialize reproducible analytical marts,
- Validate aggregation consistency across layers,
- Produce executive-ready analytical conclusions.

The system treats analytics as a structured data product rather than an exploratory notebook exercise.

2 Source Dataset: Chinook

The Chinook dataset represents a simplified digital music retail operation where customers purchase individual tracks through invoices.

Core structural characteristics:

- 412 invoices
- 2240 invoice lines

- 59 customers
- 275 artists
- 3503 tracks
- 24 billing countries
- 60 months of transactions (2021–2025)

Revenue is defined at invoice level via `invoice.total`, with line-level decomposition through `unit_price` and `quantity`.

3 Layered Architecture

The platform follows a strict three-layer architecture.

3.1 Staging Layer (stg)

Raw CSV files are ingested without transformation. This layer preserves source fidelity.

3.2 Core Layer

The core schema enforces:

- Primary keys,
- Foreign keys,
- Domain checks (non-negative monetary values),
- Referential completeness validation.

Structural validation confirmed:

No orphan invoices
 No orphan invoice lines
 No negative totals
 No missing primary keys

This guarantees structural correctness before analytical modeling.

3.3 Marts Layer

The marts layer materializes analytical models aligned with formally defined business objectives.

Implemented marts:

- `mart_revenue_time`
- `mart_market_performance`

- mart_customer_economics
- mart_portfolio_artist
- mart_portfolio_genre
- mart_revenue_composition
- mart_dependency_summary

Each mart has a defined grain, metric specification, and reconciliation rule.

4 Analytical Framework and Deliverables

The analytical logic of the platform is formalized in three integrated documents:

- **ChinookAnalytics_data_discovery.pdf** Structural and statistical profiling of the curated core schema.
- **ChinookAnalytics_analysis_objectives.pdf** Formal mapping of stakeholder questions into measurable metrics and mart definitions.
- **ChinookAnalytics_final_report.pdf** Executive-level interpretation derived from validated marts.

These documents collectively define the analytical contract of the system.

5 Key Structural Findings (Data Discovery)

The discovery phase revealed:

- Invoice totals range from 0.99 to 25.86
- Median invoice value: 3.96
- P90 invoice value: 13.86
- Quantity fixed at 1 across all invoice lines
- Revenue stable across five years (449–481 yearly range)

Identified limitations:

- No discount or promotion variable
- No basket scaling (quantity constant)
- Limited behavioral variation for churn modeling

These constraints define the boundaries of the analytical scope.

6 Analytical Results Overview

6.1 Revenue Health

- Total revenue: 2328.60
- 412 invoices across 60 months
- Revenue structurally stable across years

No strong growth or decline pattern was detected.

6.2 Market Concentration

- Top 5 countries: 58.78% of revenue
- Top 10 countries: 76.05% of revenue

Revenue is moderately concentrated geographically.

6.3 Customer Concentration

- Top 10 customers: 19.38% of revenue
- Top 20 customers: 36.61% of revenue

Customer base demonstrates relative diversification.

6.4 Catalog Distribution

- Top artist share: 5.95%
- Top 10 artists: 30.98%
- Rock genre largest contributor

Catalog risk is moderate; no single artist dominates revenue.

7 Validation Framework

A reconciliation script verifies aggregation integrity across layers.

```
core_rev      = 2328.60
time_rev      = 2328.60
market_rev    = 2328.60
customer_rev  = 2328.60
artist_rev    = 2328.60
genre_rev     = 2328.60
comp_rev      = 2328.60
dep_rev       = 2328.60
```

All marts reconcile exactly with core totals.

8 Business Interpretation and Strategic Direction

Based on the validated marts layer:

- Revenue is structurally stable but shows no clear growth trajectory.
- Geographic exposure is concentrated in a small number of countries.
- Customer dependency risk is moderate and distributed.
- Genre mix indicates dependency on Rock-driven revenue.

Strategic considerations:

- Diversify geographic exposure to reduce concentration risk.
- Expand genre mix beyond dominant segments.
- Introduce promotional or pricing experimentation to stimulate growth.
- Enrich dataset with behavioral variables (discounts, customer lifecycle).

The platform provides a validated structural view of revenue drivers and highlights concentration patterns that inform risk-aware decisions.

9 Operational Environment

The system is deployed using Docker Compose:

- PostgreSQL container
- Layered SQL execution scripts
- Deterministic pipeline execution
- Full reproducibility

Pipeline order:

`stg → core → profiling → marts → validation`

10 Conclusion

ChinookAnalytics delivers a structured analytical data platform built entirely in SQL.

The system:

- Enforces structural integrity,
- Formalizes analytical objectives,
- Produces validated analytical marts,
- Generates executive-level business insights.

The project demonstrates end-to-end analytical engineering, from ingestion to validated strategic interpretation, within a reproducible relational environment.